GALELA GAMAR MALAMU, Indonesia

WMO: **974060**

Lat: **1.8384N** Lon: **127.7880E** Elev: **16** StdP: **101.13** Time Zone: **9.00 (E09)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	20.9	21.5	19.6	14.3	24.9	20.3	15.0	24.6	6.8	28.9	5.9	28.8	0.3	N/A	0.272	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	5	7.8	33.0	26.4	32.5	26.3	32.1	26.3	27.7	31.5	27.5	31.1	27.1	30.8	3.5	N/A	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	26.7	22.3	30.3	26.3	21.9	29.9	26.1	21.5	29.6	89.0	31.7	87.3	31.3	85.9	31.0	30.6	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	6.4	5.3	4.5	DB	19.3	33.9	1.4	0.7	18.3	34.4	17.4	34.8	16.6	35.1	15.6	35.6	(4)
(5)				WB	18.5	29.0	1.7	0.9	17.2	29.6	16.2	30.1	15.2	30.6	13.9	31.3	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		26.9	26.7	26.4	26.6	27.1	27.3	26.9	26.7	26.7	26.8	27.0	27.1	27.1	(6)
(7)		DBStd		0.83	0.86	0.81	0.76	0.80	0.67	0.84	0.84	0.85	0.70	0.72	0.76	0.77	(7)
(8)		HDD10.0		0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3		0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0		6158	517	459	516	512	537	507	516	517	506	528	513	532	(10)
(11)		CDD18.3		3117	258	226	257	262	279	257	258	258	256	270	263	273	(11)
(12)		CDH23.3		28027	2244	1902	2281	2455	2683	2267	2213	2217	2296	2516	2426	2528	(12)
(13)		CDH26.7		9823	729	598	791	847	960	775	754	781	857	957	876	898	(13)
(14)	Wind	WSAvg		1.2	1.2	1.1	1.2	1.2	1.1	1.0	1.1	1.1	1.3	1.4	1.3	1.3	(14)
(15)	Precipitation	PrecAvg		2217	198	181	196	218	227	261	170	163	123	143	184	168	(15)
(16)		PrecMax		3592	438	538	871	521	396	479	420	383	259	264	367	487	(16)
(17)		PrecMin		1019	39	2	24	61	62	99	0	0	0	0	0	34	(17)
(18)		PrecStd		639	108	129	182	113	88	109	103	108	87	85	95	115	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		32.3	32.2	32.3	32.9	32.9	32.6	32.8	32.7	33.1	33.5	33.4	33.1	(19)
(20)			MCWB		26.9	26.4	26.0	26.7	26.8	26.7	26.1	26.4	25.4	24.9	27.2	27.3	(20)
(21)		2%	DB		31.7	31.4	31.6	32.0	32.2	31.9	32.0	32.0	32.3	32.6	32.2	32.1	(21)
(22)			MCWB		26.6	26.1	25.8	26.6	26.6	26.5	26.0	26.0	25.6	25.6	26.6	26.8	(22)
(23)		5%	DB		31.0	30.7	31.0	31.4	31.6	31.2	31.2	31.2	31.7	31.9	31.5	31.5	(23)
(24)			MCWB		26.3	25.8	25.6	26.3	26.4	26.1	25.8	25.7	25.7	25.8	26.2	26.5	(24)
(25)		10%	DB		30.2	30.0	30.4	30.6	30.9	30.5	30.4	30.5	30.9	31.0	30.8	30.8	(25)
(26)			MCWB		26.0	25.5	25.5	26.0	26.2	25.9	25.6	25.6	25.6	25.7	26.0	26.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		27.8	27.6	27.2	27.6	27.9	27.6	27.6	27.5	27.8	27.7	28.0	28.1	(27)
(28)			MCDB		31.3	30.7	31.0	31.2	31.4	31.3	31.5	31.2	31.2	31.5	32.2	32.1	(28)
(29)		2%	WB		27.2	26.9	26.6	27.1	27.3	27.1	26.8	26.8	27.1	27.1	27.4	27.5	(29)
(30)			MCDB		30.7	30.2	30.3	30.9	30.8	30.8	30.6	30.7	30.7	30.8	31.3	31.4	(30)
(31)		5%	WB		26.7	26.4	26.2	26.7	26.9	26.7	26.3	26.4	26.5	26.6	26.8	27.1	(31)
(32)			MCDB		30.0	29.7	29.8	30.3	30.4	30.1	30.0	30.1	30.1	30.2	30.4	30.8	(32)
(33)		10%	WB		26.3	25.9	25.8	26.3	26.5	26.2	25.9	26.0	26.0	26.1	26.3	26.5	(33)
(34)			MCDB		29.3	29.0	29.2	29.7	29.8	29.5	29.4	29.5	29.6	29.7	29.7	30.0	(34)
(35)	Mean Daily Temperature Range		MDBR		7.5	7.5	8.0	7.7	7.8	7.8	8.0	8.3	8.8	8.7	8.0	7.7	(35)
(36)		5% DB	MCDBR		8.6	8.7	9.0	8.8	8.7	8.8	9.1	9.4	10.1	10.0	9.1	8.8	(36)
(37)			MCWBR		4.0	4.1	3.9	3.8	3.9	3.9	4.0	4.1	4.5	4.1	3.9	4.0	(37)
(38)		5% WB	MCDBR		8.0	7.9	8.3	8.2	8.1	8.3	8.3	8.6	8.7	8.8	8.7	8.5	(38)
(39)			MCWBR		4.2	4.2	4.0	3.9	3.8	4.0	4.0	4.2	4.4	4.2	4.3	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.422	0.425	0.422	0.424	0.419	0.416	0.418	0.429	0.440	0.480	0.439	0.425	(40)
(41)		taud			2.485	2.464	2.477	2.466	2.481	2.500	2.496	2.451	2.408	2.276	2.408	2.468	(41)
(42)		Ebn at Noon			906	907	903	883	865	856	859	867	874	846	883	897	(42)
(43)		Edh at Noon			113	117	116	114	109	105	106	114	122	140	121	114	(43)
(44)	All-Sky Solar Radiation	RadAvg			5.00	5.36	5.62	5.69	5.39	4.90	5.11	5.49	5.78	5.77	5.36	5.08	(44)
(45)		RadStd			0.30	0.51	0.55	0.49	0.33	0.31	0.55	0.49	0.49	0.44	0.38	0.42	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page